

ECO101

Practice Sheet- Welfare & Efficiency

1. For the market of baked beans, the demand function is $P=32-8Q$ and supply function is $P=12+2Q$.
- Find the equilibrium price and quantity in this market.
 - Plot the demand and supply graph with proper labels. Denote the consumer surplus and producer surplus in your graph.
 - Calculate the total surplus.
 - Explain why a market operates efficiently at the equilibrium point.

Answer Keys:

Equilibrium Price =16

Equilibrium Quantity=2

Total Surplus = CS+ PS = 16+ 4 =20

2. Suppose the demand function is $P=200-2Q$ and supply function is $P=50+Q$. If the government introduces a price floor of $P=120$, find-
- Equilibrium price and quantity.
 - Producer Surplus and Consumer Surplus before the imposition of price floor.
 - Producer Surplus and Consumer Surplus after the imposition of price floor.
 - Deadweight loss from the price floor.

Answer Keys:

$P^*=100, Q^*=50$

Before Price Floor, CS= 2500, PS= 1250

After price floor, CS= 1600, PS=2000

DWL=150

3. Suppose for the market of green tea the demand function is $Q_d= 11000-1000P$ and supply function is $Q_s= 500P-1000$. If a price ceiling of \$4 is imposed on green tea find-
- Equilibrium price and quantity.
 - Producer Surplus and Consumer Surplus before the imposition of price ceiling.
 - Producer Surplus and Consumer Surplus after the imposition of price ceiling.
 - Deadweight loss from the price ceiling.

Answer Keys:

$P^*=8, Q^*=3000$

Before Price ceiling, CS= 4500, PS= 9000

After price ceiling, CS= 6500, PS=1000

DWL=6000

4. For market of coffee the demand function is $P=120-3Q$ and supply function is $P=30+2Q$. if the government imposes a \$15 tax on the sellers of coffee, calculate the following:
- a) New equilibrium price buyers pay, price sellers receive, and quantity sold.
 - b) New Consumer Surplus (CS) and Producer Surplus (PS) after tax.
 - c) Tax Revenue, Deadweight Loss (DWL).
 - d) Draw a properly labelled diagram based on your answers.

Answer Keys:

Quantity sold = 15

Price buyers pay = \$75

Price sellers receive = \$60

New CS = 337.5

New PS = 225

Tax Revenue = 225

Deadweight Loss = 22.5